VARIABLES DESCRIPTIONS

Prepared by: Leia Pauline S. Tonga, Research Assistant, Manila Observatory

This document lists the final sources of data followed by a description of how the following climate variables were processed: wind speed (WS), mean temperature (Tave), minimum temperature (Tmin), maximum temperature (Tmax), relative humidity (RH), heat index (HI), precipitation (PR), 3-month standard precipitation index (SPI3), 6-month standard precipitation index (SPI6), percent of normal precipitation (PNP), normalized difference vegetation index (NDVI), particulate matter with diameter less than 10μm (PM10), particulate matter with diameter less than 2.5μm (PM2P5), sulfur dioxide concentration (SO2), nitrogen dioxide concentration (NO2), carbon monoxide concentration (CO), ozone concentration (O3), ultraviolet radiation (UVR), and total solar radiation (SR).

I. DATA SOURCES

ERA5

ERA5 is a global reanalysis dataset that provides hourly (in UTC), gridded (at 0.25° x 0.25° horizontal resolution) data for atmospheric, ocean-wave and land-surface variables (further description can be found in this journal article). Among the ERA5 family of datasets, the ERA5 hourly data on pressure levels from 1940 to present and the ERA5 hourly data on single levels from 1940 to present were used to extract data for some of the needed climate variables.

The ERA5 hourly data on pressure levels from 1940 to present has a vertical coverage of 1000 hPa to 1 hPa, and a vertical resolution of 37 levels. From this, the ozone mass mixing ratio (kg kg^{-1}) and relative humidity (%) were extracted at 1000 hPa to get their surface values.

The ERA5 hourly data on single levels from 1940 to present provided the data for the following variables: 10m u-component of wind (m s^{-1}), 10m v-component of wind (m s^{-1}), 2m temperature (K), mean surface downward UV radiation flux (W m^{-2}), surface solar radiation downwards (J m^{-2}), and total precipitation (m).

- Downloading the data:
  - Create an account at Copernicus Climate Data Store
  - With an account, data is free to access and download
  - Easy download with options for spatial and temporal subsetting
- Citing the data:
  - Hersbach, H., Bell, B., Berrisford, P., Biavati, G., Horányi, A., Muñoz Sabater, J., Nicolas, J., Peubey, C., Radu, R., Rozum, I., Schepers, D., Simmons, A., Soci, C., Dee, D., Thépaut, J-N. (2023): ERA5 hourly data on pressure levels from 1940 to present. Copernicus Climate Change Service (C3S) Climate Data Store (CDS), DOI: 10.24381/cds.bd0915c6 (Accessed on DD-MMM-YYYY)
  - Hersbach, H., Bell, B., Berrisford, P., Biavati, G., Horányi, A., Muñoz Sabater, J., Nicolas, J., Peubey, C., Radu, R., Rozum, I., Schepers, D., Simmons, A., Soci, C., Dee, D., Thépaut, J-N. (2023): ERA5 hourly data on single levels from 1940 to present. Copernicus Climate Change Service (C3S) Climate Data Store (CDS), DOI: 10.24381/cds.adbb2d47 (Accessed on DD-MMM-YYYY)

CHIRPS

The Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) is a gridded rainfall dataset that combines climatological data, satellite imagery, and station data. It provides daily rainfall (mm) accumulated from 00:00Z to 23:59Z, gridded at 0.05° x 0.05° horizontal resolution (further description can be found in this journal article).

- Downloading the data:
  - Download links can be easily found at Climate Hazards Center
  - Link to CHIRPS global daily at 0.05°
  - One file per year
- Pre-processing the data: To reduce the processing time and memory needed, the data is compiled to create one dataset subsetted over the target barangays.
- Citing the data:
  - Funk, C.C., Peterson, P.J., Landsfeld, M.F., Pedreros, D.H., Verdin, J.P., Rowland, J.D., Romero, B.E., Husak, G.J., Michaelsen, J.C., and Verdin, A.P., 2014, A quasi-global precipitation time series for drought monitoring: U.S. Geological Survey Data Series 832, 4 p. http://pubs.usgs.gov/ds/832/

EAC4

The ECMWF Atmospheric Composition Reanalysis 4 (EAC4) is one of the products provided by the Copernicus Atmosphere Monitoring Service (CAMS). It is a global reanalysis dataset of atmospheric composition that provides 3-hourly (in UTC), gridded (at 0.75° x 0.75° horizontal resolution) data for the state of the atmosphere (more detailed description of the dataset can be found here). Different variables are available at different vertical coverage. For multi-level data, the surface values are obtained from model level 60.

From EAC4, the following variables were obtained: nitrogen dioxide (kg kg^{-1}), particulate matter d < 10 μm (kg m^{-3}), and particulate matter d < 2.5 μm (kg m^{-3}).

- Downloading the data:
  - Create an account at Copernicus Atmosphere Data Store
  - With an account, data is free to access and download
  - Easy download with options for spatial and temporal subsetting
- Citing the data:
  - Inness, A., Ades, M., Agustí-Panareda, A., Barré, J., Benedictow, A., Blechschmidt, A.-M., Dominguez, J. J., Engelen, R., Eskes, H., Flemming, J., Huijnen, V., Jones, L., Kipling, Z., Massart, S., Parrington, M., Peuch, V.-H., Razinger, M., Remy, S., Schulz, M., and Suttie, M.: The CAMS reanalysis of atmospheric composition, Atmos. Chem. Phys., 19, 3515–3556, https://doi.org/10.5194/acp-19-3515-2019, 2019.

MERRA-2

The Modern-Era Retrospective analysis for Research and Applications, Version 2 (MERRA-2) is a global atmospheric reanalysis dataset that incorporates satellite observations (an overview of the dataset is available here). Among the multiple collections available under MERRA-2, M2T1NXAER and M2T1NXCHM were used to extract data for some of the needed climate variables.

M2T1NXAER consists of time-averaged assimilated aerosol diagnostics at a single level including sulfur dioxide surface mass concentration. And M2T1NXCHM consists of time-averaged assimilated carbon monoxide and ozone diagnostics at a single level including carbon monoxide

surface concentration. Both gridded data are provided at a spatial resolution of 0.5° x 0.625° and a temporal resolution of 1-hour.

- Downloading the data:
    - Create an account at Earthdata Login and go to GES DISC
    - With an account, data is free to access and download
    - Easy download with options for spatial and temporal subsetting
- Citing the data:
    - Global Modeling and Assimilation Office (GMAO) (2015), MERRA-2 tavg1_2d_aer_Nx: 2d,1-Hourly,Time-averaged,Single-Level,Assimilation,Aerosol Diagnostics V5.12.4, Greenbelt, MD, USA, Goddard Earth Sciences Data and Information Services Center (GES DISC), Accessed: [Data Access Date], 10.5067/KLICLTZ8EM9D
    - Global Modeling and Assimilation Office (GMAO) (2015), MERRA-2 tavg1_2d_chm_Nx: 2d,1-Hourly,Time-Averaged,Single-Level,Assimilation,Carbon Monoxide and Ozone Diagnostics V5.12.4, Greenbelt, MD, USA, Goddard Earth Sciences Data and Information Services Center (GES DISC), Accessed: [Data Access Date], 10.5067/3RQ5YS674DGQ

MAIAC

Among the available data products, the MODIS/Terra+Aqua Vegetation Index from Multi-Angle Implementation of Atmospheric Correction (MAIAC), Daily L3 Global 0.05 Deg CMG (MCD19A3CMG) collection is selected as the source for NDVI. The dataset provides multiple variables but the "NDVI_gapfill" is extracted to represent daily NDVI, as it is the most stable according to the developer.

- Downloading the data:
    - Create an account at Earthdata Login and go to LP DAAC
    - Data is downloaded from Index of /MOTA/MCD19A3CMG.061
    - One folder per year and in each folder, one file per day
- Pre-processing the data: To reduce the processing time and memory needed, the data is compiled to create one dataset subsetted over the target barangays.
- Citing the data:
    - Lyapustin, A. (2023). MODIS/Terra+Aqua Vegetation Index from MAIAC, Daily L3 Global 0.05Deg CMG V061 [Data set]. NASA EOSDIS Land Processes Distributed Active Archive Center. Accessed [Data Access Date] from https://doi.org/10.5067/MODIS/MCD19A3CMG.061
- Correspondence with the USGS EROS User Services regarding the MCD19 Product
    - From the PI (Alexei Lyapustin) - "One current problem with these data is lack of geometric normalization. The NDVI depends on view geometry – e.g. NDVI is higher for the forward scattering and is lower for the backscattering view geometry. Over dense vegetation, the difference can be ~0.02-0.05 and higher over urbanized surfaces. When I update different parts of [the] image from different orbits (different view geometries), this create[s] a patchy pattern although the variation is not high."
    - The exchange provided an insight on the current limitations of the dataset.

II. CLIMATE VARIABLES

Wind Speed (WS) [m s^{-1}]

The wind speed is derived from the eastward and northward components of surface winds. ERA5 provided the hourly u- and v-components of 10m winds used to calculate WS following this formula: $WS = (WS_x^2 + WS_y^2)^{½}$ where WS_x and WS_y are the u- and v-components, respectively.

UTC was converted to PHT by adding 8 hours to the time. The daily values of WS_x and WS_y were determined by taking the average per day (00:00 PHT - 23:00 PHT). Afterwards, the daily WS was calculated.

Mean, Minimum and Maximum Temperature (Tave, Tmin, Tmax) [°C]

The daily mean, minimum and maximum temperatures are derived from the 2m temperature data from ERA5. First, UTC was converted to PHT by adding 8 hours to the time. Tave was determined by taking the average temperature per day (00:00 PHT - 23:00 PHT). Tmin was determined by taking the minimum temperature per day (00:00 PHT - 23:00 PHT). And Tmax was determined by taking the maximum temperature per day (00:00 PHT - 23:00 PHT). These values in Kelvin were converted to degrees Celsius by subtracting 273.15.

Note that ERA5 was selected as the source of data for temperature after validation with observed station data.

Relative Humidity (RH) [%]

Relative humidity is available in ERA5 at different pressure levels, where surface values are at 1000 hPa. The hourly RH in UTC was converted to PHT by adding 8 hours to the time. The daily values were determined by taking the average per day (00:00 PHT - 23:00 PHT).

Heat Index (HI) [°C]

Heat index, also known as apparent temperature, is derived from the air temperature and relative humidity. To calculate the HI, a MetPy implementation was used. The 2m temperature (K) and relative humidity at 1000 hPa (%) from ERA5 were used as input. But first, UTC was converted to PHT by adding 8 hours of time. After implementing MetPy, the daily mean HI is calculated. The default unit of the output is in °F, but it was converted to °C.

Precipitation (PR) [mm/day]

Precipitation is extracted from CHIRPS, which provides total precipitation accumulated from 00:00Z to 23:59Z. PR was extracted per location and no further processing was performed.

Note that CHIRPS is a land-only dataset. Among the target barangays, Barangay Tumalutab (PH097332095) in Zamboanga City, has no available data. This barangay is a small island isolated from the rest of the province. It was not represented as land in the dataset.

Additionally, CHIRPS was selected as the source of data for precipitation after validation with observed station data.

3- and 6-month Standardized Precipitation Index (SPI3, SPI6) [dimensionless]

The Standardized Precipitation Index (SPI) is a meteorological drought index derived from standardizing precipitation deviations from the long-term climatic norm. SPI calculation involves several steps. First, the long-term monthly precipitation data (at least 30 years) for a specific

location is collected. SPI3 is accumulated for over three months, while SPI6 for over six months. Then, the precipitation data is fitted to a probability distribution function (PDF), typically the gamma distribution. Based on the PDF, the cumulative distribution function (CDF) is calculated, representing the probability that precipitation falls below a certain threshold. The probabilities obtained from the CDF are transformed into standard normal variates, representing the SPI values. Positive SPI values indicate wetter conditions than average, while negative values suggest drier conditions. (Reference: Standardized Precipitation Index User Guide (WMO-No. 1090))

The daily precipitation per target barangay from 1991-2022 is extracted from CHIRPS. For SPI3 calculation, the 3-monthly accumulated precipitation is calculated (i.e. no values for January 1991 and February 1991, the value for March 1991 is given by the sum of the daily precipitation values from 01 January 1991 to 31 March 1991, ..., the value for December 2022 is given by the sum of the daily precipitation values from 01 October 2022 to 31 December 2022). And for SPI6 calculation, the 6-monthly accumulated precipitation is calculated (i.e. no values for January to May 1991, the value for June 1991 is given by the sum of the daily precipitation values from 01 January 1991 to 30 June 1991, ..., the value for December 2022 is given by the sum of the daily precipitation values from 01 July 2022 to 31 December 2022). The resulting data is subsetted from 2003 to 2022, thus, there is monthly SPI3 and SPI6 value for the target barangays except for Brgy. Tumalutab, Zamboanga City because there is no precipitation data available.

- SPI values and interpretations:
    - >= 2.0 Extremely wet
    - 1.5 to 1.99 Very wet
    - 1.0 to 1.49 Moderately wet
    - -0.99 to 0.99 Near normal
    - -1.0 to -1.49 Moderately dry
    - -1.5 to -1.99 Severely dry
    - <= -2.0 Extremely dry
- But the interpretation of SPI values must be done with caution especially for short accumulation periods. One must account for the climatological normal in a specific area. In an area where rainfall is normally low during a 3-month period, large negative or positive SPIs may result even though the departure from the mean is relatively small.
    - In the case of Dagupan City, -inf/inf values are expected since it normally has very low rainfall values for the 1-, 3-month periods during the dry season, which is basically comparing the difference from normal to zero (climatologically) i.e. dividing by zero.
        - The calculation for SPI involves taking the log of the climatological normal. If the climatological normal is zero, then the code will give a warning that a divide by zero is encountered. This results in the -inf/inf values.
        - Issues with Infinity SPI Values: sbegueria says that “sometimes there is no solution to the distribution fitting problem and the SPEI / SPI yields NaN or Inf. The problem is not in the code, but in the data. Sometimes the data does not conform to the distribution. Maybe there [is] too [little] data, the data consists of mostly zeros, or other weird issues.”

Percent of Normal Precipitation (PNP) [%]

The percent of normal precipitation (PNP) is a simple measurement of rainfall for a location, derived by dividing actual precipitation by the climatological mean (or normal) precipitation and

multiplying by 100%. It can be calculated for a variety of time scales, ranging from a single month to a group of months representing a specific season, to an annual or water year. (Reference: Percent of Normal Precipitation)

The PNP calculation is done using the following formula: PNP = 100% x (Pi / P) where Pi is the actual precipitation of the considered month, and P is the mean long-term (1991-2020) precipitation for each month, or monthly normal.

PNP is prone to misunderstanding because it gives different indications of conditions depending on the location and season. Therefore, in the data, there are two additional columns with the "PNP" column, namely, "PR_TOT" (accumulated rainfall for a given month) and "PR_NORM" (accumulated monthly normal rainfall for the period of 1991-2020). There are instances of minimal precipitation in a location with a large PNP. This occurs for areas with almost zero PR_NORM.

Note that the precipitation values are derived from CHIRPS, which is a land-only dataset. Since the grid for Brgy Tumalutab, Zamboanga City is not included as part of land in the CHIRPS dataset, there is no rainfall data for the location and thus, there are also no values for PNP.

Normalized Difference Vegetation Index (NDVI) [dimensionless]

The Normalized Difference Vegetation Index (NDVI) quantifies vegetation health and density based on satellite or aerial imagery. It operates on the principle that healthy vegetation reflects more near-infrared (NIR) light and absorbs more red light.

NDVI is calculated using the following formula:

$$NDVI = \frac{(NIR-Red)}{(NIR+Red)},$$

where NIR is the near-infrared reflectance and Red is the red reflectance.

Its values range from -1 to +1, with higher positive values indicating healthier and more abundant vegetation, and values close to zero indicating sparse or stressed vegetation. Negative values often correspond to non-vegetated surfaces such as water bodies or urban areas.

- References:
  - Rouse, J. W., Haas, R. H., Schell, J. A., & Deering, D. W. (1973). Monitoring vegetation systems in the Great Plains with ERTS. In Third Earth Resources Technology Satellite-1 Symposium-Volume I: Technical Presentations (pp. 309-317).
    - MONITORING VEGETATION SYSTEMS IN THE GREAT PLAINS WITH ERTS
  - Tucker, C. J. (1979). Red and photographic infrared linear combinations for monitoring vegetation. Remote Sensing of Environment, 8(2), 127-150.
    - Red and Photographic Infrared Linear Combinations for Monitoring Vegetation

The daily NDVI is extracted from the MAIAC dataset, specifically the gap-filled NDVI variable. Before extraction, the daily files are processed to create one gridded dataset. This dataset has "NDVI_gapfill" values (valid range: 0-10000) with the scale factor (0.0001) already applied. No additional processing is done during the extraction of NDVI data per location per day. Generally, NDVI has a range of -1 to +1, however, the dataset only provides 0 and positive values, because the MAIAC method only considers land-only and land-containing ocean tiles. The final table of extracted values has complete daily data for all locations except for Brgy. Tumalutab, Zamboanga City, as it is represented as an ocean tile in MCD19A3CMG.

Particulate Matter d < 10 µm and < 2.5 µm (PM10, PM2P5) [µg m^{-3}]

The surface concentration of particulate matter with diameter less than 10 µm (PM_{10}) and less than 2.5 µm ($PM_{2.5}$) are extracted from EAC4. Both variables are provided at a single level. The data is in units of kilograms per cubic meter. (The variable descriptions for PM_{10} and $PM_{2.5}$ are found here). The conversion to units of microgram per cubic meter was done using the following formula: PM_d in µg m^{-3} = PM_d in kg m^{-3} * 10^9.

EAC4 provides 3-hourly data in UTC, which was converted to PHT by adding 8 hours to the time. The daily values for both PM10 and PM2P5 are calculated by taking the mean of the values from 00:00 PHT to 21:00 PHT. The first available data in EAC4 is on 2003-01-02 00:00 UTC. Thus, in local time, there is data from 2003-01-02 08:00. As such, there is no value for 2003-01-01, and the value for 2003-01-02 is the mean of the available 6 timesteps. For 2003-01-03 onwards, there is complete data, already in PHT, for all timesteps.

Sulfur Dioxide Concentration (SO2) [ppbv]

The surface concentration of sulfur dioxide is given by the SO2SMASS (SO2 Surface Mass Concentration) variable from M2T1NXAER of MERRA-2. The data is provided in kilograms per cubic meter units and is converted to parts per billion units using the following formula:

Concentration (ppm) = 24.45 x concentration (mg m^{-3}) ÷ molecular weight

SO2 in ppbv = 24.45 * (SO2 in kg m^{-3})*(10^9 mg kg^{-1}) / 64.06 g mol^{-1}

(Reference)

M2T1NXAER is provided in UTC, which was converted to PHT by adding 8 hours to the time. The daily values were calculated by taking the mean SO2SMASS values from 00:00 PHT to 23:00 PHT. To quantify the reliability of each daily data, the number of available hourly time steps are counted and stored in an additional column, “N_TIMESTEPS_AGGREGATED”.

Nitrogen Dioxide Concentration (NO2) [ppbv]

The surface concentration of nitrogen dioxide is extracted from EAC4 at model level 60. The data is in units of kilograms (of nitrogen dioxide) per kilogram (of air). (The variable description for NO_2 can be found here). The conversion to units of parts per billion using the following equation: NO2 in ppbv = (28.9644 / 46.0055) * 10^9 * NO2 in kg kg^{-1} (Reference)

EAC4 provides 3-hourly data in UTC, which was converted to PHT by adding 8 hours to the time. The daily value is calculated by taking the mean NO2 from 00:00 PHT to 21:00 PHT. The first available data in EAC4 is on 2003-01-02 00:00 UTC. Thus, in local time, there is data from 2003-01-02 08:00. As such, there is no value for 2003-01-01, and the value for 2003-01-02 is the mean of the available 6 timesteps. For 2003-01-03 onwards, there is complete data, already in PHT, for all timesteps.

Carbon Monoxide Concentration (CO) [ppmv]

The surface concentration of carbon monoxide is given by the COSC (CO Surface Concentration, ppmv) variable from M2T1NXCHM of MERRA-2. The data is provided in parts per million units and is converted to parts per parts per billion using the following formula:

COSC in ppmv = COSC in ppbv * 10^{-3}

M2T1NXCHM is provided in UTC, which was converted to PHT by adding 8 hours to the time. The daily values were calculated by taking the mean COSC values from 00:00 PHT to 23:00

PHT. To quantify the reliability of each daily data, the number of available hourly time steps are counted and stored in an additional column, "N_TIMESTEPS_AGGREGATED".

Ozone Concentration (O3) [ppbv]

The ozone concentration is extracted from ERA5, which has a parameter for ozone mixing ratio available at different levels. Surface values of O3 are given by the values at 1000 hPa. (Parameter details are found here). The data is in kilograms (of ozone) per kilogram (of air), and converted to parts per billion using the following equation: O3 in ppbv = (28.9644 / 47.9982) * 10^9 * O3 in kg kg^{-1} (Reference)

The hourly O3 data in UTC was converted to PHT by adding 8 hours to the time, and the daily values were determined by taking the average per day (00:00 PHT - 23:00 PHT).

Ultraviolet Radiation (UVR) [W m^{-2}]

There are many radiation variables in ERA5 available at a single level. Among these, the mean surface downward UV radiation flux (W m^{-2}) is selected to represent the ultraviolet radiation. The parameter is described as the amount of ultraviolet (UV) radiation reaching the surface or passing through a horizontal plane. According to the documentation, the average energy flux for a day is derived from the individual fluxes for each hour from 1:00 to 23:00 of that day and 0:00 of the next day.

To account for the 8 hour difference between UTC and PHT and the averaging to be done from 01:00 UTC to the next day 00:00 UTC, 7 hours was added to the original timestamp. Then, the mean of the individual fluxes from 00:00 PHT to 23:00 PHT was calculated to get the daily value.

Total Solar Radiation (SR) [W m^{-2}]

There are many radiation variables in ERA5 available at a single level. Among these, the surface solar radiation downwards (SSRD, J m^{-2}) is selected to represent the total solar radiation. The parameter is described as the total amount of direct and diffuse solar radiation that reaches a horizontal plane at the surface of the earth. Converting from J m^{-2} to W m^{-2} and calculating for the daily average follows the documentation, where the individual fluxes from 1:00 PHT to 23:00 PHT and 0:00 of the next day are accumulated, and divided by 86400 s to get the daily average.

To account for the 8 hour difference between UTC and PHT and the averaging to be done from 01:00 UTC to the next day 00:00 UTC, 7 hours was added to the original timestamp.